

LIMSLite

Complete Step-by-Step Testing Guide

Pharma LIMS Testing Guide

This guide provides a complete, sequential walkthrough for testing all modules of the LIMSLite Pharma Laboratory Information Management System. Follow each step in order to validate the full workflow from Master Data Setup through Batch Release.

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Final Verification Checklist

PHASE 0 — Master Data Setup

Complete all 3 steps below before registering any samples. These steps configure the system foundation.

Step 1: Login as Admin

Navigate to the LIMSLite login page. Enter the Admin credentials to access system configuration.

Field	Value
URL	http://localhost:3000 (or your deployed URL)
Username	admin
Password	Admin@123
Role	System Administrator
Expected Landing Page	Admin Dashboard / Master Data menu

E-SIGNATURE Password: Admin@123 | Meaning: Login Confirmation | Reason: Authenticate as System Admin

Step 2: Create Laboratory

Navigate to Master Data > Laboratories > Create New. Enter the details below and save.

Field	Value
Laboratory Name	APEX Quality Control Lab
Laboratory Code	APEX
Laboratory Type	QC
Location / Site	Building A, Floor 2
Accreditation	ISO 17025:2017
Status	Active

E-SIGNATURE Password: Admin@123 | Meaning: Record Creation | Reason: Create APEX QC Laboratory

Step 3: Create Material

Navigate to Master Data > Materials > Create New. Enter Paracetamol API details.

Field	Value
Material Name	Paracetamol API
Material Code	PARA-001
Material Type	API (Active Pharmaceutical Ingredient)
CAS Number	103-90-2
Shelf Life	730 days (24 months)
Storage Condition	Store below 25°C, protect from moisture
Unit of Measure	kg
Manufacturer	PharmaChem Ltd.
Status	Active

E-SIGNATURE Password: Admin@123 | Meaning: Record Creation | Reason: Create Paracetamol API Material

Step 4: Create Sample Types

Navigate to Master Data > Sample Types > Create New. Create both sample types below.

Type Name	Code	Category	Physical State	Status
Finished Product	FP	Finished	Solid	Active
Raw Material	RM	Incoming	Solid	Active

E-SIGNATURE Password: Admin@123 | Meaning: Record Creation | Reason: Create Sample Types FP and RM

Step 5: Create Parameters

Navigate to Master Data > Parameters > Create New. Create all three parameters listed below.

Parameter Name	Code	Unit	Data Type	Critical	Status
Assay%	ASSAY	%	Numeric	YES	Active
Moisture Content	MC	%	Numeric	No	Active
pH	PH	-	Numeric	No	Active

Note: For Assay%, set Min Decimal Places = 1, Max Decimal Places = 2

E-SIGNATURE Password: Admin@123 | Meaning: Record Creation | Reason: Create Analytical Parameters

Step 6: Create Specification Template

Navigate to Master Data > Specifications > Create New Template. Use USP compendial standard.

Field	Value
Template Name	PARA-FP-SPEC-001
Version	v1.0
Compendial Standard	USP (United States Pharmacopeia)
Material	Paracetamol API (PARA-001)
Sample Type	Finished Product (FP)
Status	Active / Approved

Specification Limits:

Parameter	Code	Min Limit	Max Limit	Unit	Method
Assay%	ASSAY	98.0	102.0	%	HPLC-USP
Moisture Content	MC	0.0	0.5	%	KF-USP
pH	PH	4.5	6.5	-	PH-USP

E-SIGNATURE Password: Admin@123 | Meaning: Template Approval | Reason: Approve Specification Template PARA-FP-SPEC-001 v1.0

Step 7: Create Form Template

Navigate to Master Data > Form Templates > Create New. This template defines the Electronic Batch Record (EBR) form for Finished Product testing.

Field	Value
Form Name	Standard FP Testing Form
Form Code	FT-FP-001
Applicable Sample Type	Finished Product
Version	v1.0
Form Type	TimeBased
Evidence Collection	Mandatory
E-Signature Required	Yes — Analyst + Reviewer
Linked Specification	PARA-FP-SPEC-001 v1.0
Status	Active

Note: TimeBased forms require analysts to record actual start and end time for each test section.

E-SIGNATURE Password: Admin@123 | Meaning: Template Activation | Reason: Activate Form Template FT-FP-001 v1.0

Step 8: Add Instrument

Navigate to Master Data > Instruments > Add New. Register the HPLC instrument used for Assay testing.

Field	Value
Instrument ID	HPLC-001
Instrument Name	Agilent HPLC System
Instrument Type	HPLC
Manufacturer	Agilent Technologies
Model	1260 Infinity II
Serial Number	AGIL-1260-2024-0042
Location	APEX QC Lab — Room 204
Status	Available
Calibration Due Date	31-Dec-2026
Last Calibration	01-Jan-2026

E-SIGNATURE Password: Admin@123 | Meaning: Instrument Registration | Reason: Register HPLC-001 in the system

Step 9: Create Users

Navigate to Admin > User Management > Create User. Create the three operational users below. Each will be used in subsequent phases.

Full Name	Username	Password	Role	Lab Assignment
Dr. Priya Nair	priya.nair	Priya@123	Analyst	APEX QC Lab
Mr. Rajan Mehta	rajan.mehta	Rajan@123	QCLead	APEX QC Lab
Dr. Sarah QA	sarah.qa	Sarah@123	QA	APEX QC Lab

Role Permissions Summary:

Role	Register Sample	Test Results	Review Results	COA Approval	Batch Release
Analyst	Yes	Yes	No	No	No
QCLead	Yes	Yes	Yes	No	No
QA	No	No	Yes	Yes	Yes

Note: Set email addresses for each user to enable notification workflows.

E-SIGNATURE Password: Admin@123 | Meaning: User Account Creation | Reason: Create Analyst, QCLead, and QA user accounts

Step 9B: Configure Environmental Checkpoint (Optional)

Navigate to Master Data > Checkpoints > Create New. This sets up routine lab environment monitoring.

Field	Value
Checkpoint ID	CP-ENV-001
Checkpoint Type	TimeBased
Schedule Times	08:00, 14:00, 20:00 (3x daily)
Parameter Monitored	pH (lab purified water)
Location	APEX QC Lab — Utility Point
Assigned To	On-duty Analyst

PHASE 1 — Sample Registration

Register the Paracetamol API Finished Product sample and assign it to the work queue for testing.

Step 10: Register a Sample

Login as QCLead (rajan.mehta). Navigate to Sample Management > Register New Sample. Complete all fields:

Field	Value
Sample ID (auto-generated)	APEX-FP-20260602-0001
Lot / Batch Number	LOT-PARA-2026-001
Material	Paracetamol API (PARA-001)
Sample Type	Finished Product (FP)
Collection Date	02-Jun-2026
Expiry Date	02-Jun-2028 (730 days)
Laboratory	APEX Quality Control Lab
Quantity Collected	50 g
Collected By	Mr. Rajan Mehta
Storage Location	Cold Room A — Shelf 3
Priority	Normal
Specification Template	PARA-FP-SPEC-001 v1.0
Form Template	FT-FP-001 v1.0

E-SIGNATURE Password: Rajan@123 | Meaning: Sample Registration | Reason: Register Paracetamol API FP sample LOT-PARA-2026-001

Step 11: Assign Sample to Work Queue

After registration, navigate to Work Queue > Assign. Select the sample and assign to Dr. Priya Nair (Analyst).

Field	Value
Sample ID	APEX-FP-20260602-0001
Assign To (Analyst)	Dr. Priya Nair (priya.nair)
Testing Due Date	04-Jun-2026
Priority	Normal
Instructions	Run full panel: ASSAY, MC, pH per USP methods
Instruments Needed	HPLC-001, KF Titrator, pH Meter
Expected Turnaround	48 hours

E-SIGNATURE Password: Rajan@123 | Meaning: Work Assignment | Reason: Assign sample APEX-FP-20260602-0001 to Analyst priya.nair

PHASE 2 — Work Queue & Testing

Analyst opens the EBR form, performs testing using calibrated instruments, enters results, and submits for review.

Step 12: Login as Analyst

Logout of QCLead session. Login with Analyst credentials.

Field	Value
Username	priya.nair
Password	Priya@123
Role	Analyst
Expected Landing Page	My Work Queue dashboard

E-SIGNATURE Password: Priya@123 | Meaning: Login Confirmation | Reason: Analyst login to access work queue

Step 13: View Work Queue

Navigate to Work Queue. Confirm the assigned sample appears with status "Pending Testing".

Field	Value
Sample ID	APEX-FP-20260602-0001
Status	Pending Testing
Due Date	04-Jun-2026
Assigned By	Mr. Rajan Mehta
Material	Paracetamol API (PARA-001)
Form Template	FT-FP-001 v1.0

Note: If the sample is not visible, verify the lab assignment matches "APEX QC Lab".

Step 14: Start Testing / Open EBR Form

Click "Start Testing" on the sample row. The Electronic Batch Record (EBR) form FT-FP-001 will open. Record the following:

Field	Value
Test Start Date/Time	02-Jun-2026 09:00
Analyst Name	Dr. Priya Nair
Instrument Used (HPLC)	HPLC-001 — Agilent HPLC System
Instrument Calibration Status	Valid — Cal Due 31-Dec-2026
Environmental Conditions	Temp: 22°C, Humidity: 45%
Sample Condition	Normal — No damage observed
SOP Reference	SOP-QC-HPLC-001 Rev 3

E-SIGNATURE Password: Priya@123 | Meaning: Test Initiation | Reason: Start testing EBR for APEX-FP-20260602-0001

Step 15: Enter Test Results

In the EBR form, enter the measured values for each parameter. The system will auto-evaluate pass/fail against PARA-FP-SPEC-001 limits.

Parameter	Code	Measured Value	Unit	Min Limit	Max Limit	Status
Assay%	ASSAY	100.5	%	98.0	102.0	PASS
Moisture Content	MC	0.3	%	0.0	0.5	PASS
pH	PH	5.2	-	4.5	6.5	PASS

Evidence / Attachments:

Evidence Type	Description	File / Reference
Chromatogram	HPLC Assay raw data	HPLC-001-20260602-PARA.pdf
Instrument Log	HPLC-001 run log	HPLC001-log-20260602.xlsx
Raw Data Sheet	Moisture & pH handwritten	RAW-FP-20260602-0001.pdf

Note: Upload scanned evidence files before submitting. Mandatory per FT-FP-001 form settings.

Step 16: Submit Results for Review

After entering all results and uploading evidence, click "Submit for Review". Record completion details:

Field	Value
Test End Date/Time	02-Jun-2026 14:30
Testing Duration	5 hours 30 minutes
Overall Status (auto)	PASS — All 3 parameters within specification
Submit To	Mr. Rajan Mehta (QCLead)
Comments	All tests completed per SOP-QC-HPLC-001 Rev 3. No deviations observed.

E-SIGNATURE Password: Priya@123 | Meaning: Results Submission | Reason: Submit test results for APEX-FP-20260602-0001 to QC Lead for review

PHASE 3 — Results Review

QC Lead reviews submitted test results, verifies data integrity and approves or returns for correction.

Step 17: Login as QC Lead / Reviewer

Logout of Analyst session. Login with QC Lead credentials to access the review queue.

Field	Value
Username	rajan.mehta
Password	Rajan@123
Role	QCLead
Expected Landing Page	QC Dashboard — Pending Review items

E-SIGNATURE Password: Rajan@123 | Meaning: Login Confirmation | Reason: QC Lead login to access review queue

Step 18: Review & Approve Results

Navigate to Results Review Queue. Locate APEX-FP-20260602-0001. Click "Review". Verify each result entry:

Check Item	Analyst Value	Specification	Review Action	Outcome
Assay% result	100.5%	98.0–102.0%	Verified against chromatogram	Approved
Moisture result	0.3%	0.0–0.5%	Cross-checked with raw data	Approved
pH result	5.2	4.5–6.5	Checked pH meter log	Approved
Evidence files	3 files uploaded	All mandatory	Downloaded and reviewed	Complete
Instrument status	HPLC-001	Calibration valid	Verified Cal Due date	Approved

Field	Value
Review Decision	Approved — Proceed to COA Generation
Reviewer Comments	All results verified against raw data and instrument logs. Data integrity confirmed.
Review Date/Time	03-Jun-2026 10:15
Reviewed By	Mr. Rajan Mehta (QCLead)

E-SIGNATURE Password: Rajan@123 | Meaning: Results Approval | Reason: Approve test results for APEX-FP-20260602-0001 — all parameters PASS

PHASE 4 — COA Review & Approval

Generate the Certificate of Analysis, review all entries, and obtain QA approval signature.

Step 19: Generate Certificate of Analysis (COA)

Login as QC Lead. Navigate to Sample APEX-FP-20260602-0001 > Actions > Generate COA. The system will auto-populate the following:

Field	Value
COA Number (auto)	COA-2026-00001
Sample ID	APEX-FP-20260602-0001
Lot / Batch Number	LOT-PARA-2026-001
Material	Paracetamol API (PARA-001)
Sample Type	Finished Product
Testing Laboratory	APEX Quality Control Lab
Specification Reference	PARA-FP-SPEC-001 v1.0 (USP)
Date of Analysis	02-Jun-2026
Date of COA Generation	03-Jun-2026
Overall Result	COMPLIES (All parameters within specification)
Analyst	Dr. Priya Nair
Reviewed By	Mr. Rajan Mehta (QCLead)

COA — Results Summary Section:

Test Parameter	Specification Limit	Reported Result	Compliance
Assay (by HPLC)	98.0 – 102.0 %	100.5 %	PASS
Moisture Content	Not more than 0.5 %	0.3 %	PASS
pH	4.5 – 6.5	5.2	PASS

E-SIGNATURE Password: Rajan@123 | Meaning: COA Generation | Reason: Generate COA-2026-00001 for LOT-PARA-2026-001

Step 20: Review & Approve COA

Send the COA to QA (Dr. Sarah QA) for final review. QA logs in and navigates to COA Approval queue.

Field	Value
QA Login	sarah.qa / Sarah@123
Navigate To	QA Dashboard > COA Approval Queue
COA to Review	COA-2026-00001
Review Checklist	Verify Lot No., Material, Results, Spec version, Analyst & Reviewer signatures
COA Approval Decision	Approved
QA Approval Date/Time	03-Jun-2026 14:00
Approved By	Dr. Sarah QA (QA)

E-SIGNATURE Password: Sarah@123 | Meaning: COA Approval | Reason: Approve COA-2026-00001 for LOT-PARA-2026-001
Paracetamol API

PHASE 5 — Batch Release

QA performs final batch disposition review and formally releases the batch for use or distribution.

Step 21: QA Final Review

QA (Dr. Sarah QA) is already logged in. Navigate to Batch Release > Pending Release. Locate LOT-PARA-2026-001.

Field	Value
QA Username	sarah.qa
Navigate To	Batch Release > Pending Release
Lot Number	LOT-PARA-2026-001
Material	Paracetamol API (PARA-001)
COA Number	COA-2026-00001
COA Status	Approved (03-Jun-2026)
All Test Results	PASS (Assay: 100.5%, Moisture: 0.3%, pH: 5.2)
Deviations / OOS	None
Review of Deviations	Not Applicable — No deviations raised
GMP Documentation Review	Complete
Final Batch Disposition	RELEASED

Note: If any deviation or OOS result exists, it must be formally closed before release.

Step 22: Release Batch

Click "Release Batch" button. Enter QA credentials for e-signature. The system will update batch status to RELEASED and timestamp the event.

Field	Value
Batch / Lot Number	LOT-PARA-2026-001
Sample ID	APEX-FP-20260602-0001
COA Number	COA-2026-00001
Release Decision	RELEASED
Release Date/Time	03-Jun-2026 15:30
Released By	Dr. Sarah QA (QA)
Final Status in LIMS	BATCH RELEASED
Post-Release Actions	COA available for download; batch record locked (read-only)
Notification Sent To	rajan.mehta, priya.nair (system notification)

E-SIGNATURE Password: Sarah@123 | Meaning: Batch Release Authorization | Reason: Release LOT-PARA-2026-001 — Paracetamol API Finished Product

Final Verification Checklist

After completing all 22 steps, verify each item below. Tick the checkbox and record the verifier name and date.

Phase 0 — Master Data Setup		
☐ Admin login verified	Verified by:	Date:
☐ Laboratory APEX created and active	Verified by:	Date:
☐ Material Paracetamol API (PARA-001) created	Verified by:	Date:
☐ Sample Types FP and RM created	Verified by:	Date:
☐ Parameters ASSAY, MC, PH created	Verified by:	Date:
☐ Specification Template PARA-FP-SPEC-001 v1.0 created and approved	Verified by:	Date:
☐ Form Template FT-FP-001 v1.0 created and active	Verified by:	Date:
☐ Instrument HPLC-001 registered, calibration valid	Verified by:	Date:
☐ Users priya.nair, rajan.mehta, sarah.qa created with correct roles	Verified by:	Date:
Phase 1 — Sample Registration		
☐ Sample APEX-FP-20260602-0001 registered (LOT-PARA-2026-001)	Verified by:	Date:
☐ Sample assigned to Analyst priya.nair in work queue	Verified by:	Date:
Phase 2 — Work Queue & Testing		
☐ Analyst priya.nair logged in and viewed work queue	Verified by:	Date:
☐ Sample status shows "Pending Testing" in work queue	Verified by:	Date:
☐ EBR form FT-FP-001 opened successfully	Verified by:	Date:
☐ All 3 test results entered: Assay 100.5%, Moisture 0.3%, pH 5.2	Verified by:	Date:
☐ All 3 results auto-evaluated as PASS by the system	Verified by:	Date:
☐ Evidence files (3) uploaded and attached	Verified by:	Date:
☐ Results submitted for review to QC Lead	Verified by:	Date:
Phase 3 — Results Review		
☐ QC Lead rajan.mehta logged in and found sample in review queue	Verified by:	Date:
☐ All results verified against raw data and evidence; approved	Verified by:	Date:
Phase 4 — COA Review & Approval		
☐ COA COA-2026-00001 generated successfully with all results	Verified by:	Date:
☐ QA sarah.qa reviewed and approved COA-2026-00001	Verified by:	Date:
Phase 5 — Batch Release		
☐ QA final review completed — no deviations or OOS	Verified by:	Date:
☐ Batch LOT-PARA-2026-001 released; status = BATCH RELEASED	Verified by:	Date:
☐ COA available for download; batch record locked (read-only)	Verified by:	Date:
☐ System notification sent to analysts	Verified by:	Date:

Test Execution Sign-Off		
Analyst Dr. Priya Nair	QC Lead / Reviewer Mr. Rajan Mehta	QA Authorised Signatory Dr. Sarah QA
Signature / Date	Signature / Date	Signature / Date

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